

Naive Bayes Classifier

only works for classification Problem. Classify based Probability. Date:

It assumes that all features are independent from each other.

Conditional Probability:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad \text{where } P(B) > 0$$

Probability of event A happening given that event B has already happened.

$P(A \cap B) \rightarrow$ Probability of A and B happen together

$P(B) \rightarrow$ Probability of event B.

Bayes Theorem:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \rightarrow \textcircled{1}$$

$$P(B|A) = \frac{P(B \cap A)}{P(A)} = \frac{P(A \cap B)}{P(A)} \rightarrow \textcircled{2} \quad [A \cap B = B \cap A]$$

from eqn ① and ② combining $P(A \cap B)$

$$P(A|B) P(B) = P(B|A) P(A)$$

$$\Rightarrow \boxed{P(A|B) = \frac{P(B|A) P(A)}{P(B)}} \quad \text{where } P(B) > 0$$

If B can occur for multiply mutually exclusive event $A_1, A_2, \dots, A_n$ then for k-th A event

$$P(A_k|B) = \frac{P(B|A_k) P(A_k)}{\sum_{i=1}^n P(B|A_i) \cdot P(A_i)}$$

If $A_1, A_2, \dots, A_n$ form a partition of the sample space (mutually exclusive and exhaustive) then

$$\begin{aligned} P(B) &= \sum_{i=1}^n P(B \cap A_i) \\ &= \sum_{i=1}^n P(B|A_i) P(A_i) \end{aligned}$$

$$\left[\begin{aligned} P(B|A) &= \frac{P(A \cap B)}{P(B)} \\ \Rightarrow P(A \cap B) &= P(B|A) P(B) \end{aligned} \right]$$

Naive Bayes assume that all features are independent given the class.

Suppose $X = (x_1, x_2, \dots, x_n)$ is the input feature vectors.

C is the label. (set of possible classes).

we want to find $P(C|X) = \frac{P(X|C) \cdot P(C)}{P(X)}$

$P(C) \rightarrow$ probability of class C

$P(X|C) \rightarrow$ likelihood of seeing feature X given class C .

$P(X) \rightarrow$ evidence. as it is same for all the classes so often ignored.

so prediction is done by

$$\hat{C} = \underset{C}{\operatorname{argmax}} P(C|X) \propto P(C) \cdot P(X|C)$$

naive assumption:

$$P(X|C) = P(x_1, x_2, \dots, x_n|C)$$

$$\approx \prod_{i=1}^n P(x_i|C)$$

If dataset $X = \{x_1, x_2, x_3, \dots, x_n\} \{y\}$

so according to the bayes theorem we can say we have to find out the probability of class y when feature x is given.

$$P(y|x_1, x_2, \dots, x_n) = \frac{P(x_1, x_2, \dots, x_n|y) P(y)}{P(x_1, x_2, \dots, x_n)}$$

$$= \frac{P(x_1|y) \cdot P(x_2|y) \cdot P(x_3|y) \cdot \dots \cdot P(x_n|y) \cdot P(y)}{P(x_1) \cdot P(x_2) \cdot \dots \cdot P(x_n)}$$

$$= \frac{P(y) \cdot \prod_{i=1}^n P(x_i|y)}{\prod_{i=1}^n P(x_i)}$$

$\rightarrow$ ignore this part cause it will be same for every class.

$$P(y|x_1, x_2, \dots, x_n) \propto P(y) \cdot \prod_{i=1}^n P(x_i|y)$$

$$\hat{y} = \underset{y}{\operatorname{argmax}} P(y) \cdot \prod_{i=1}^n P(x_i|y)$$

return which y has the highest probability

For categorical feature:

• If there are K unique values and in our dataset there are n classes then make a crosstab table of $K \times n$ shape.
then for every unique value find the probability of this feature to be in n -th class.

temp feature

	yes	no
High	2	5
low	3	4
	5	9

$$P(\text{High}|\text{yes}) = \frac{2}{5} \quad P(\text{low}|\text{yes}) = \frac{3}{5}$$

and so on

If there is any 0 at any cell of a feature then add 1 to the table for all features.

For numerical feature: Assume that feature is normally distributed.
 $N(\mu, \sigma)$. find mean and std.

Now instead of counting the frequency use the probability density function (PDF).

for a feature X with mean μ and standard deviation σ in class $Y=y$. The probability of observing a value x is

$$P(X=x | Y=y) = \frac{1}{\sqrt{2\pi\sigma_y^2}} \cdot e^{-\frac{(x-\mu_y)^2}{2\sigma_y^2}}$$

where

$\mu_y \rightarrow$ mean of feature X for class y

$$= \frac{1}{\sigma_y \sqrt{2\pi}} \cdot e^{-\frac{1}{2} \left(\frac{x-\mu_y}{\sigma_y} \right)^2}$$

$\sigma_y \rightarrow$ std of feature x for class y

Binomial Naive Bayes:

Binomial distribution: a random variable $X \sim \text{Binomial}(n, p)$

$$P(X=k) = \binom{n}{k} \cdot p^k \cdot (1-p)^{n-k}$$

$n \rightarrow$ num of trial (fixed)

$k \rightarrow$ observed value (num of success)

$p \rightarrow$ Probability of a single trial.

Let' we have a feature that counts success (like num +ve word in a document) then the conditional probability with binomial PMF

$$P(X=k | Y=y) = \binom{n}{k} p_y^k (1-p_y)^{n-k}$$

where p_y is estimated from the train data for class y .

$$p_y = \frac{\text{total success in class } y}{\text{no. of trials.} \rightarrow n \times \text{number of sample in class } y}$$

Bernoulli is a special case of Binomial where $n=1$.

If $n=1$ (only one trial)

k can be 0 (fail) or 1 (success)

Bernoulli Naive Bayes:

• each feature is binary 0/1.

• For a class $Y=y$ we want the probability that a feature X is 1

$$P(X=1 | Y=y) = \frac{\text{\#times feature=1 in class } y + \alpha}{\text{\#samples in class } y + 2\alpha}$$

count $\rightarrow$

$\uparrow$ laplace smoothing

some times a feature might never be 1 in class $y \rightarrow \text{Probability} = 0$

so by multiplying 0 with other feature prob will make zero. to avoid we add α .

Gaussian Naive Bayes

Dataset $D = \{ (x^{(i)}, y^{(i)}) \}_{i=1}^N$ where $x^{(i)} \in \mathbb{R}^d$ (d numeric features) Date :
 $y^{(i)} \in \{1, 2, \dots, k\}$

Naive Bayes assumption: features are conditionally independent.

$$P(x|y) = \prod_{j=1}^d P(x_j|y)$$

Gaussian assumption: for each class y and each feature index j ,

$$P(x_j|y) = \mathcal{N}(x_j | \mu_{y,j}, \sigma_{y,j}^2)$$

$\uparrow$ mean of j -th feature index for class y

$\xrightarrow{\text{variance}}$

At training phase:

From the training set find class priors, mean and variance.

Prior (MLE):

$$\hat{\pi}_y = P(y) = \frac{N_y}{N}$$

Maximum Likelihood

Estimation

• Feature mean (per class):

$$\hat{\mu}_{y,j} = \frac{1}{N_y} \sum_{i: y^{(i)}=y} x_j^{(i)}$$

$\uparrow$ j -th feature

• Feature variance (per class): 2 choices

MLE (population variance)

$$\hat{\sigma}_{y,j}^2 = \frac{1}{N_y} \sum_{i: y^{(i)}=y} (x_j^{(i)} - \hat{\mu}_{y,j})^2$$

unbiased same variance

$$s_{y,j}^2 = \frac{1}{N_y - 1} \sum_{i: y^{(i)}=y} (x_j^{(i)} - \hat{\mu}_{y,j})^2$$

[Note] $\hat{\sigma}_{y,j}^2 = 0$ then add 10^{-9} to avoid zero division error.



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Likelihood for a test point (Product of Gaussians)

given a new observation $x = (x_1, x_2, \dots, x_d)$

The class condition likelihood

$$\begin{aligned} P(X=x|y=y) &= \prod_{j=1}^d P(x_j|y) \\ &= \prod_{j=1}^d \frac{1}{\sigma_{yj} \sqrt{2\pi}} \times e^{-\frac{1}{2} \left(\frac{x_j - \mu_{yj}}{\sigma_{yj}} \right)^2} \end{aligned}$$

now posterior (Baye's Rule)

$$P(Y=y|X=x) = \frac{P(X=x|Y=y) P(y)}{\sum_{y'=1}^K P(x|y') \cdot P(y')}$$

now classification: x belong to the class y' for which $P(y|x)$ is max.

$$\hat{y} = \arg \max_y P(Y=y|X=x)$$

or we can say $\hat{y} = \arg \max_y \underbrace{P(X=x|Y=y) P(y)}_{\text{PDF}}$

multiplying many small PDF with $P(y)$ can go underflow. So take log of them.

Log-unnormalized Posterior (log-score) for class y :

$$S_y(x) = \log P(y) + \log P(X=x|Y=y)$$

$$= \log P(y) + \log \prod_{j=1}^d N(x_j | \mu_{yj}, \sigma_{yj}^2)$$

$$= \log P(y) + \prod_{j=1}^d \log \left[\frac{1}{\sqrt{2\pi} \sigma_{yj}} \times e^{-\frac{1}{2} \left(\frac{x_j - \mu_{yj}}{\sigma_{yj}} \right)^2} \right]$$

$$S_y(x) = \log P(y) - \frac{1}{2} \sum_{j=1}^d \log(2\pi \sigma_{yj}^2) - \frac{1}{2} \sum_{j=1}^d \left(\frac{x_j - \mu_{yj}}{\sigma_{yj}} \right)^2$$

Decision Rule:

$$\hat{y} = \arg \max_y S_y(x)$$

Date :

Algorithm Steps:

Training:

① For each class y : compute $\hat{\pi}_y = \frac{N_y}{N}$

② For each feature j and class y : Compute $\hat{\mu}_{yj}$ and $\hat{\sigma}_{yj}^2$
add ϵ if $\hat{\sigma}_{yj}^2$ is 0

Prediction for sample x

① For each class y compute

$$S_y(x) = \log \hat{\pi}_y - \frac{1}{2} \sum_{j=1}^d \log(2\pi \hat{\sigma}_{yj}^2) - \frac{1}{2} \sum_{j=1}^d \left(\frac{(x_j - \hat{\mu}_{yj})^2}{\hat{\sigma}_{yj}^2} \right)$$

② Predict $\hat{y} = \arg \max_y S_y(x)$

log posterior $\rightarrow$

$$S_y(x) = \underbrace{\log \hat{\pi}_y}_{\text{log prior}} + \underbrace{\left[-\frac{1}{2} \sum_{j=1}^d \log(2\pi \hat{\sigma}_{yj}^2) \right]}_{\text{normalization}} + \underbrace{\left[-\sum_{j=1}^d \frac{(x_j - \hat{\mu}_{yj})^2}{2 \hat{\sigma}_{yj}^2} \right]}_{\text{exponent}}$$

gaussian log pdf



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Naïve Bayes for mix feature (Numerical + Categorical)

naïve Bayes relies on the conditional independence assumption.

$$P(y | x_1, x_2, \dots, x_d) \propto P(y) \prod_{j=1}^d P(x_j | y)$$

now for numerical and categorical feature

$$P(y | x_{\text{num}}, x_{\text{cat}}) \propto P(y) * \prod_{j \in \text{num}} P(x_j | y) * \prod_{k \in \text{cat}} P(x_k | y)$$

For categorical feature use frequency count

$$P(x_k = v | y) = \frac{\text{count of samples } x_k = v \text{ and class } y + \alpha}{\text{count of samples with class } y + \alpha \cdot n_v}$$

$x_k \rightarrow k$ - the categorical feature

$x_k = v \rightarrow x_k$ may have multiple unique value we are calculating for value v . ex: x_k is Outlook then v can be Overcast, Sunny.

$n_v \rightarrow$ number of unique values in x_k column.

For numerical feature use gaussian PDF to find $P(x_j | x_{\text{num}})$

$$P(x_j | y) = \frac{1}{\sqrt{2\pi} \sigma_{y,j}} \times e^{-\frac{1}{2} \times \frac{(x_j - \mu_{y,j})^2}{\sigma_{y,j}^2}}$$

now combine both

$$\begin{aligned} \log P(y | x) &= \log P(y) + \sum_{j \in \text{num}} \log P(x_j | y) + \sum_{j \in \text{cat}} \log P(x_j | y) \\ S_y(x) &= \underbrace{\log P(y)}_{\text{Log Prior}} + \underbrace{\sum_{j \in \text{num}} \left[-\frac{1}{2} \log(2\pi \sigma_{y,j}^2) - \frac{(x_j - \mu_{y,j})^2}{2 \sigma_{y,j}^2} \right]}_{\text{numerical (Gaussian)}} + \underbrace{\sum_{k \in \text{cat}} \log \left(\frac{\text{count}(x_k = v, y) + \alpha}{\text{count}(y) + \alpha n_v} \right)}_{\text{Categorical (smoothed freq)}} \end{aligned}$$